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Improved automated detection of glaucoma by correlating fundus and SD-OCT image analysis

Tehmina Shehryar¹  | Muhammad U. Akram² | Samina Khalid¹ | Shamila Nasreen¹ | Anum Tariq¹ | Aqib Perwaiz² | Arslan Shaukat²

¹Department of Software Engineering, Mirpur University of Science & Technology, Mirpur, Pakistan

²Department of Computer Engineering, College of E&ME, NUST, Islamabad, Pakistan

Correspondence

Tehmina Shehryar, Department of Software Engineering, Mirpur University of Science & Technology, Mirpur 10240, Pakistan.

Email: tehmينا.se@must.edu.pk

Abstract

Glaucoma is a multifactorial ocular disease. Ophthalmologists mostly use fundus or optical coherence tomography (OCT) for diagnosis of glaucoma. In this study, a hybrid computer-aided-diagnosis (H-CAD) system has been proposed that integrates both fundus and OCT imaging technologies for reliable diagnosis of glaucoma. Fundus module inspects the outer layer of eye's posterior part. It considers a variety of structural and textural features and makes a decision using support vector machine (SVM). In OCT module, the cup to disc ratio (CDR) has been computed by examining the internal layers of the retina. The cup contour has been extracted from inner-limiting-membrane (ILM) layer using a set of novel techniques for the calculation of cup diameter. Similarly, in the disc diameter calculation the retinal-pigment-epithelium (RPE) layer termination points have been identified by a number of innovative strategies to locate disc margin. Furthermore, a new criterion based on the mean value of RPE-layer end points has been proposed for the determination of cup edges. A local-dataset annotated by four ophthalmologists has been used for evaluation of proposed H-CAD system. The evaluations and results have shown that the final result of H_CAD system is more trustable than its contemporary automated models.

KEYWORDS

biomedical informatics, telemedicine, computer aided diagnosis, computer vision, fundus, glaucoma, optical coherence tomography

1 | INTRODUCTION

Glaucoma is a chronic neurological disease. It is one of the preeminent causes of irreversible blindness. Currently, the estimated worldwide patients of glaucoma are in excess of 64 million.¹ It is an asymptomatic disease in initial and intermediate stages which reveals its symptoms only in the advanced and untreatable stage; when the eyesight is confined only to the central vision, also known as “tunnel vision”.² The absence of prior warning in this disease has contributed to the frightening records of undiagnosed glaucoma patients all over the world.

Almost 50% of all the glaucoma patients remains undiagnosed until disease reaches at the last and incurable stage even in medically advanced countries like the USA and Australia.^{3,4} In Pakistan, there are more than 1.8 million glaucoma patients and again half of them have already lost vision permanently.⁵ This high percentage of undiagnosed patients indicates the need for substantial changes and improvement in existing diagnosis procedures. Prediagnosis is commonly done manually using tonometry to measure intra-ocular pressure (IOP) and gonioscopy to inspect anterior chamber of the eye. These examinations are usually followed by optic nerve

head (ONH) analysis. The glaucoma is symptomless and painless disease, therefore it is important for people especially over 30 years of age to get eyes regularly examined to avoid blindness. In order to perform mass screening, the ophthalmologists have to screen a lot of patients. One of the cost-effective solutions to this problem is to develop CAD systems using modern imaging technologies. Automated screening systems can reduce time and cost of the screening process. It can also decrease the workload on ophthalmologists as they only have to inspect those cases which are diagnosed by CAD as potentially diseased cases. There is a variety of CAD systems developed for the diagnosis of glaucoma; however, the results show the room for improvement. The main research gap is the lack of a complete system which can accurately detect glaucoma at an early and treatable stage.

The ophthalmologists mostly use fundoscopy or OCT to diagnose glaucoma. Funduscopy gives an enlarged view of posterior part of the eye. It can capture structural changes on the outermost layer of the retina.⁶ On the other hand, the OCT imaging technology has the ability to create objective and reproducible measurements of topographic parameters of internal retinal layers. It functions as a form of "Optical biopsy".⁷ Structural changes in internal layers of retina occur prior to visual field defects. In-time treatment of glaucoma can prevent 98% of the vision. In this study, we present a reliable and robust automated system that combines advantages of both fundus and OCT imaging modalities to detect glaucoma at an early and treatable stage. It considers a variety of glaucoma condition using both fundus and OCT image of the same patient. A unique combination of structural and textural features including a novel superpixel algorithm has been used for classification of fundus images using SVM classifier. Similarly, a set of novel strategies has been adopted for accurate computation of cup and disc diameter using OCT images. Since cup to disc ratio (CDR) is a prime clinical indicator for the diagnosis of glaucoma; therefore, it has been used as an indicator in the analysis of both types of images.⁸ The cup is the central white portion of OD with no nerve fibers. In glaucoma, decay of fiber nerves cause cup size to increase and as a result CDR

also increases.⁹ Figure 1 shows both OCT and fundus images of OD portion of left eye of a glaucomatous patient. Figure 1A shows OCT image of left eye of a glaucoma patient where, cup is marked with yellow lines and disc is marked with green lines. Likewise, Figure 1B shows fundus image of same glaucomatous patient with central white portion in blue rectangle as cup and outer bigger blue square shows disc. The green and yellow lines indicated disc and cup diameter respectively. The CDR value less than or equal to 0.5 is considered as normal whereas greater than 0.5 is considered as glaucomatous. The final outcome after fusion of both modules has given improved results with 96% accuracy and 100% sensitivity.

2 | RELATED WORK

In a recent work, Maheshwari et al,¹⁰ have involved the red, green, blue, and gray scale image of colored fundus image to extract texture features by applying two-dimensional empirical wavelet transform using a two-dimensional (2D) empirical little wood Paley wavelet transform. The correntropy has been extracted from the decomposed components of 2D EWT. Features with the highest value t value have been used for classification using SVM. The system attained 96.6% accuracy on a local dataset of 60 images. Similarly, in the same year, Acharya et al,¹¹ presented another solution for automated diagnosis of glaucoma using textural features. The adaptive histogram equalization technique has been used to enhance the contrast of gray scale image. The Leung-Malik (LM), Schmid (S), and maximum response (MR4 and MR8) filter banks have been applied to the gray-scale images and textons have been extracted. Local configuration pattern (LCP) has been used to extract required spectral and phase features from the textons. The sequential forward search method followed by t -test helps in feature selection and ranking. Salient parameters are fed to different classifiers for discrimination. An accuracy of 95.7%, specificity of 93.7%, and sensitivity of 96.2% is achieved using KNN classifier and LM filter bank for six significant features on a local dataset of 703 images. In Reference 12,

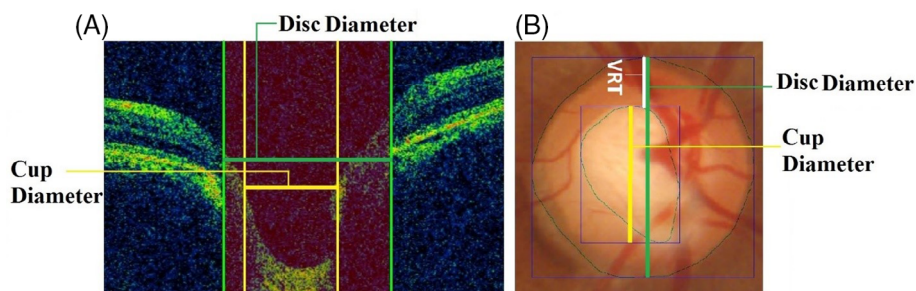


FIGURE 1 OCT and fundus image of OD portion of left eye of a patient. (A) SD-OCT image of OD portion. (B) Fundus image of OD portion. Yellow line shows Cup diameter and Green line shows Disc diameter [Color figure can be viewed at wileyonlinelibrary.com]

the structural feature CDR has been considered for automated detection of glaucoma. The ROI has been extracting by selecting region with largest area with highest brightness as optic disc. The green channel has been used for segmentation of cup. Median filter has been applied to remove noise. The Biography Based Optimization and Kapur's test have been used to find out threshold points of each fundus image. Adaptive threshold technique have been used for segmentation. Eight neighbors connectivity have been applied to get accurate shape of cup. Finally, area of both extracted cup and disc have been calculated and CDR value greater than 0.3 has been considered as glaucoma and less than 0.3 has been taken as normal cases. The system attained 97.58% accuracy on a local dataset of 290 images. In another research, the statistical features have been extracted from fundus image and used for classification using SVM classifier and achieved 93.10% accuracy.¹³

Likewise, the CDR has been computed from OCT images in a recent work.¹⁴ The disc margins have been calculated using neuro canal opening (NCO) on retinal-pigment-epithelium (RPE) layer. The cup border has been identified on inner-limiting-membrane (ILM) layer considering NCO location. In Reference 15, the disc diameter has been computed based on the distance between RPE layer termination points. The cup diameter has been calculated at a distance of 150 μm from the optic disc edges on ILM layer. In another research, both the fundus and OCT images have been involved in the diagnosis of glaucoma.¹⁶ In fundus images, CDR was calculated using different segmentations algorithms for both cup and disc segmentation. In OCT images, RPE layer and Retinal Nerve Fiber (RNF) layer have been used to define reference plane to calculate cup and disc diameter. Results analysis revealed superior performance of OCT images in the diagnosis of glaucoma. Also, previously in 2012, Retinal Vitreal (RV) and Retinal Choroid (RC) layers have been used to detect cup and disc boundaries using OCT images and gained 92% accuracy using 125 images.¹⁷ Chen et al¹⁸ formulate a hybrid feature set by fusing structural feature CDR computed from fundus images with textural features extracted from OCT images. The gcForest technique was used for the classification of glaucoma. The proposed system achieved 96.88% accuracy. Marzieh et al examined the symmetry between both eyes by correlating the CDR values.¹⁹ The OCT images has been used for analysis. The ILM and RPE layer has been used to detect cup and disc. The ratio of summed areas of cup to summed areas of disc has been computed to local CDRs. Forty healthy OCT datasets of size $650 \times 512 \times 128$ (attained from Topcon 3D OCT-1000) and corresponding 1536×1612 fundus images has been used for evaluation of system. Maximum symmetry was observed in middle of cup and disc regions. In a recent study, deep learning method using capsule network for glaucoma diagnosis has been used.²⁰ The system was trained on three-dimensional

(3D) OCT-labeled images without any region or structural segmentation. The proposed network was accessed on 50 volumes and achieved 0.97 for the area under curve the ROC curve.

3 | PROPOSED METHODOLOGY

The proposed hybrid computer-aided-diagnosis (H-CAD) system has two main modules, that is, fundus and OCT modules. The fundus module analyses the changes caused by glaucoma on the outer layer of eye's posterior part. The analysis of fundus images has been done in two distinct ways. In first method, three structural features, that is, rim-to-disc ratio (RDR), CDR, and cup shape have been extracted from optic disc. In the second method, 115 textural and intensity based features have been extracted by assessing the overall appearance and texture of sharply cropped ONH part of the fundus image. Texture features extract a unique image pattern from image which can be effectively used for classification of glaucoma. In both methods of fundus module analysis, the SVM classifier has been used for the classification. Whereas, in OCT module, spectral domain OCT (SD-OCT) images have been used to capture changes induced by glaucoma to the internal layers of the retina. The vertical cup and disc diameter has been calculated by identifying cup and disc margin by using the internal RPE layer of the retina. The vertical CDR is then computed and the sample is labeled as diseased or normal on the basis of the CDR value. The proposed system relies on triple synthesis of three diverse systems for the decision making. Primarily, we rely on the analysis of OCT module, which is highly sophisticated and most modern method of diagnosing glaucoma. The two fundus modules assist in the decision making. When a case is diagnosed positive by using OCT module and one of the fundus modules also provides the same result then the case is confirmed as glaucoma. However, if none of the fundus modules conform to the findings of OCT module, then the case is treated as a suspected one. Likewise, the case is treated as normal if an OCT module fails to detect glaucoma and one or both the fundus modules match the result provided by an OCT module. Figure 2 shows a flow diagram of proposed H-CAD system for screening of glaucoma.

3.1 | Fundus module

In fundus module, bilinear interpolation has been used to enhance spatial resolution and to bring all images on same scale of 1150×1150 .²¹ Optic nerve analysis is required for detection of glaucoma. The proposed system uses intensity information along with blood vessel density to locate OD from fundus.²²

The process of OD localization starts with preprocessing. In the preprocessing stage, back ground estimation and

FIGURE 2 Flow diagram for proposed system [Color figure can be viewed at wileyonlinelibrary.com]

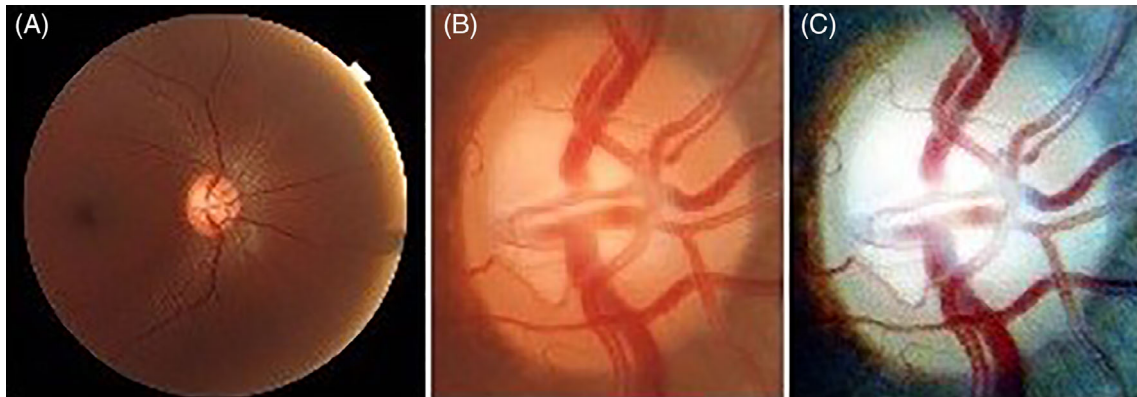
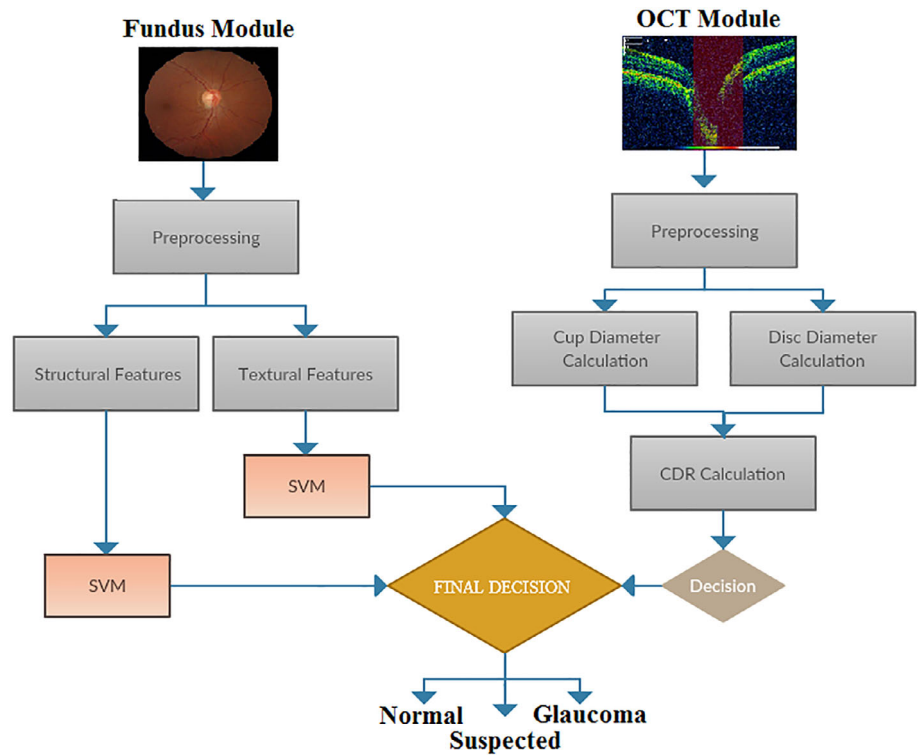


FIGURE 3 Enhanced and cropped image, (A) original image, (B) cropped image, and (C) E_n image [Color figure can be viewed at wileyonlinelibrary.com]

noise removal has been done. In next stage, candidate OD regions are detected followed by blood vessel detection. Finally, OD detection has been done on the base of vessel density in the candidate OD region. The contrast of cropped portion has been enhanced using histogram equalization technique.²³ Figure 3B,C shows cropped and enhanced (E_n) images, respectively.

3.1.1 | Structural features

Cup to disc ratio(f_1)

In glaucoma, the cup size increases due to the decay of fiber nerves causing a significant difference between the

CDR value of glaucomatous and a healthy eye. In the proposed system, the CDR value less than 0.5 is considered as normal while greater than 0.5 is considered as glaucomatous.²⁴ Figure 4 describes the complete process of proposed CDR calculation. The enhanced image (E_n) has been used for disc segmentation (Figure 4B). An empirically calculated threshold value $T = 0.4$ has been applied on red channel of E_n . The morphological opening with a disk-shaped structuring element with a radius of 15 has been applied on binary image after thresholding to remove and isolate irrelevant noisy object. Any noisy object having size smaller than predefined value of disc size (DS) has been further removed. The value of DS has been selected based on average size of disc in given

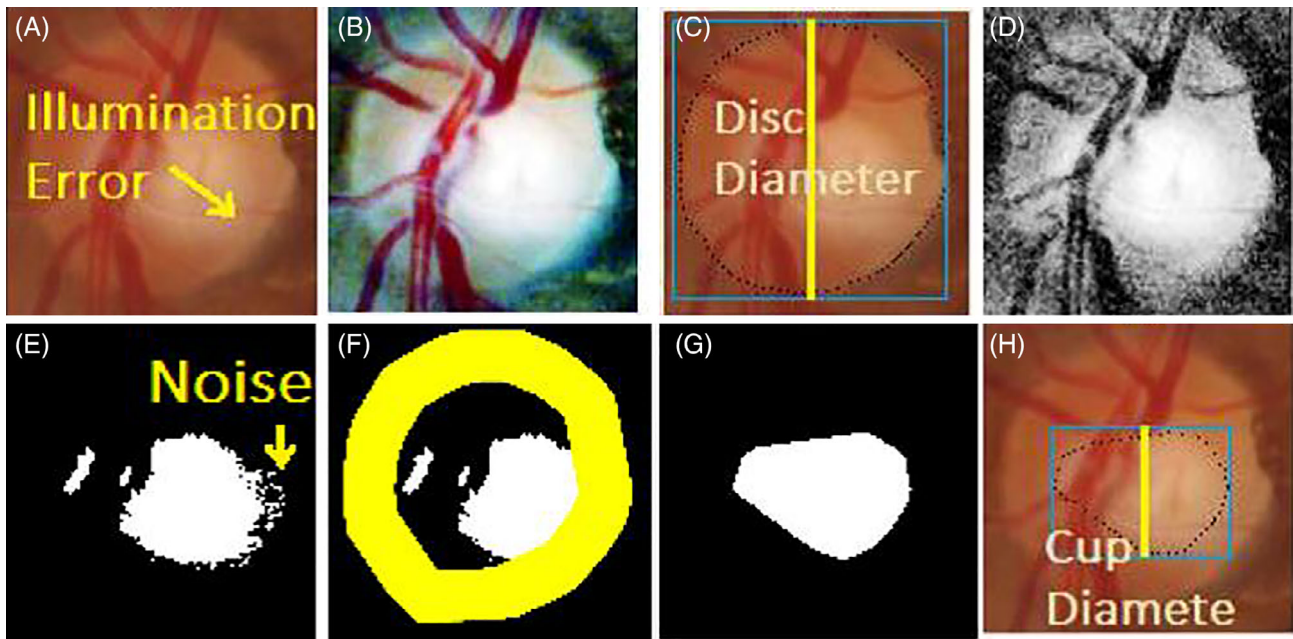


FIGURE 4 Vertical cup diameter calculation, (A) OD Image. The arrow points illumination noise, (B) E image, (C) vertical disc diameter calculated using proposed system, (D) blue channel of E image, (E) thresholding blue channel. Noise as a result of illumination error can be seen after thresholding, (F) illumination noise removal based on disc edges, (G) morphological closing and computing convex hull, (H) vertical cup diameter [Color figure can be viewed at wileyonlinelibrary.com]

dataset. Convex hull has been computed to create a polygon which can contain the entire disc.

The Canny edge detector has been used to identify boundary of disc and using “regionprops” method the vertical disc diameter has been computed. The “regionprops” method measure properties of eight connected components in the binary image. Figure 4C shows the vertical disc diameter calculated using proposed method. Likewise, in cup diameter calculation process the blue channel of E_n image has been used for cup segmentation (Figure 4D). An empirically calculated threshold value $T = 0.9$ has been used to segment cup as shown in Figure 4E. It can be perceived that the resultant binary image has segmented all the bright pixels including those bright pixels which have been formed due to illuminations (In some cases, false bright pixels may occur due to peripapillary atrophy [PPA]). We have termed such bright pixels as “false bright pixels” in cup segmentation process. The formation of false bright pixels causes error in cup extraction. In the proposed system, false bright pixels have been catered by specifying a region based on disc edges as “Noise” area. Any object in this region is considered as “false bright pixels” and removed accordingly. Figure 4F shows “Noise” area highlighted in the yellow. The MIN filter has been used to remove salt noise.

The morphological closing using a large disk-shaped structuring element with a radius of 50 has been applied to close all remaining bright pixels to

form cup. Furthermore, objects smaller than cup size (CS) is also removed. Where CS is derived from average cup size value in given dataset. Convex hull has been computed to merge all cup portion in to a single cup (Figure 4G). It can be seen in Figure 4G, that the proposed technique has removed the false bright pixels from the binary image (Figure 4F). The Canny edge detector has been applied to extract boundary of cup, while the cup vertical diameter has been computed using regionprops” method as shown in Figure 4H. The CDR has been computed by taking ratio of cup vertical diameter (CVD) to the disc vertical diameter (DVD) as described in Equation (1).

$$CDR = \frac{CVD}{DVD} \quad (1)$$

The Table 1 shows results of $T = 0.1$ to $T = 0.9$ on red and blue channel of input image. It can be observed from the table data that in each case best fit has been obtained on $T = 0.4$ for disc segmentation using red channel and $T = 0.9$ for cup segmentation using blue channel.

Vertical RDR calculation (f_2)

The area between cup and disc boundary is known as rim. In the proposed system, only the vertical rim thickness (VRT) is computed from the already extracted cup and disc boundary. VRT is the wideness of the superior portion of

TABLE 1 Results of $T = 0.1$ to $T = 0.9$ using red and blue channels

Input Image	Results of $T=0.1$ to $T=0.9$									
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Red Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Blue Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Red Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Blue Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Red Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Blue Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Red Channel
	$T = 0.1$	$T = 0.2$	$T = 0.3$	$T = 0.4$	$T = 0.5$	$T = 0.6$	$T = 0.7$	$T = 0.8$	$T = 0.9$	Blue Channel

rim only and it is calculated by taking the difference of upper y-coordinates of cup and disc bounding box. In Figure 1B, the white line shows VRT on the OD. Vertical rim thinning is one of the signs of glaucoma.²⁵ Finally, RDR has been calculated by taking the ratio of VRT to the DVD as described in Equation (2). The RDR value less than 0.1 indicates glaucomatous eye.²⁶

$$\text{RDR} = \frac{\text{VNRR}}{\text{DVD}} \quad (2)$$

Cup shape analysis (f_3)

Due to the movement of blood vessels, the cup shape changes in the glaucomatous eye. As compared to the glaucoma eye, the cup counter is more irregular in the normal eye due to the presence of blood vessels.²⁷ Figure 5B-1,B-2

shows cup contour of a normal and glaucoma eye respectively. In proposed system, fractal analysis of cup contour has been done using Hausdorff's fractal dimension.²⁸ The cup contour has been extracted by subtracting eroded cup from initially extracted cup. The Hausdorff's fractal dimension can be computed using following equation, where ϵ is the Euclidean dimension of the object.

$$D_0 = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{\log \epsilon^{-1}} \quad (3)$$

The fractal dimension values found to be in the range of 1.11 to 1.15 for the glaucoma eye and the range was 1.16 to 1.25 for the normal eye during the analysis of the given dataset.

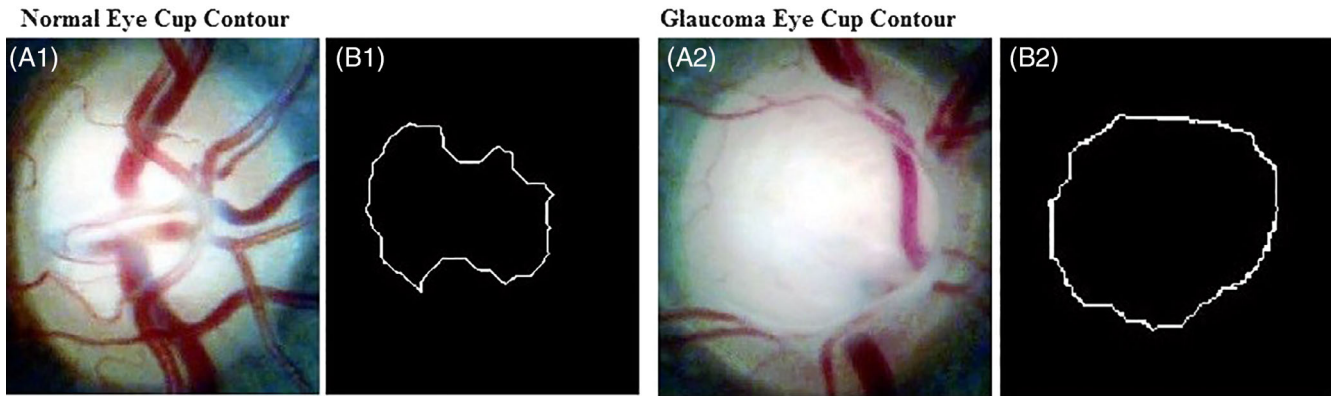


FIGURE 5 Cup contour, (A-1) OD of a normal eye, (B-1) extracted cup contour of the normal eye, (A-2) OD of the glaucomatous eye, and (B-2) extracted cup contour of the glaucomatous eye [Color figure can be viewed at wileyonlinelibrary.com]

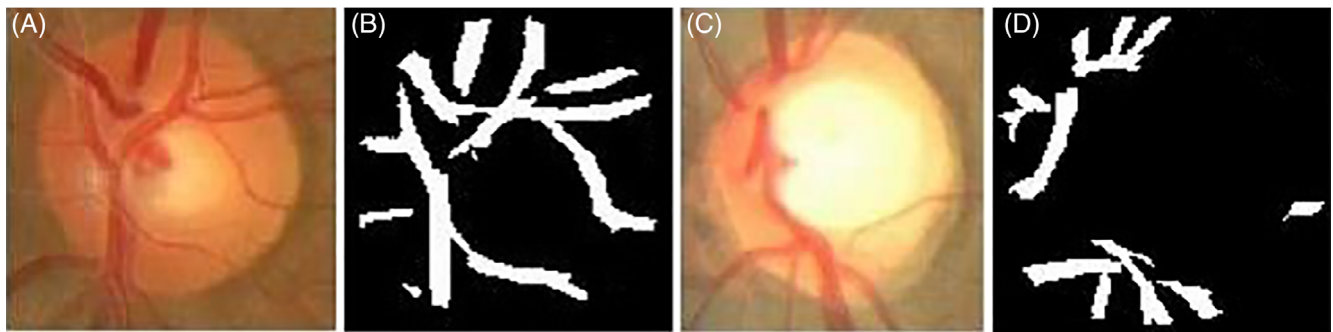


FIGURE 6 Rim Vessels Extraction, (a) normal eye OD, (b) rim vessels in normal eye, (c) glaucoma eye OD, (d) rim vessels in glaucoma eye [Color figure can be viewed at wileyonlinelibrary.com]

3.1.2 | Textural features

Rim vessels extraction (f_4 - f_9)

The vascular changes can also be very helpful for the early diagnosis of glaucoma.²⁹ In the proposed system, initially the blood vessels from the entire OD portion have been segmented using multilayered thresholding based blood vessel segmentation as proposed in Reference 30. A rim mask derived from the already calculated cup and disc boundary is then applied on extracted blood vessels to segment rim vessels only. Figure 6 shows extracted rim vessels using proposed method. Six textural features including the total vessels area (f_4), kurtosis of the distribution of vessels areas (f_5), largest region area in vessel map (f_6), smallest region area in vessels map (f_7), mean of areas in the region of vessel map (f_8), and SD on the region of the areas distribution on vessel map (f_9) has been calculated.

Gray level co-occurrence matrix (f_{10} - f_{14})

The statistics computed by gray level co-occurrence matrix (GLCM) features can be used to capture the excavation occurred by glaucoma in OD of a glaucomatous eye. In the proposed system, following five GLCM features have been used for diagnosis of glaucoma.³¹

$$\text{Contrast} = \sum_{i=1}^G \sum_{j=1}^G (i-j)^2 P(i,j) \quad (4)$$

$$\text{Entropy} = \sum_{i=1}^G \sum_{j=1}^G P(i,j) \times \log_2(P(i,j)) \quad (5)$$

$$\text{Correlation} = \sum_{i=1}^G \sum_{j=1}^G \frac{\{i \times j\} \times P(i,j) - \{\mu_x \times \mu_y\}}{\sigma_x \times \sigma_y} \quad (6)$$

$$\text{Cluster shade} = \sum_{i=1}^G \sum_{j=1}^G \{i + j - \mu_x - \mu_y\}^3 \times P(i,j) \quad (7)$$

$$\text{Energy} = \sum_{i=1}^G \sum_{j=1}^G (P(i,j))^2 \quad (8)$$

In the above equations, $P(i, j)$ is the gray level co-occurrence matrix. G denotes the number of gray levels used and μ_x , σ_x , μ_y , σ_y are means and standard deviations of P_x and P_y .

Discrete wavelet transform (f_{14} - f_{54})

The discrete wavelet transform (DWT) can be used to analyze frequency variations in horizontal, vertical and diagonal orientations. It can be used to isolate smooth variations in an image. In proposed system, the low pass filter has been used to down sample the input image by a factor of two.³² The input image has been decomposed to level four using “coiflet” discrete wavelet. The “coiflet” discrete wavelet has better capabilities for image smoothing and edges removal.³³ Finally, the first two distributions moments of each column of decomposed image has been computed.

Mean gray-level (f_{55})

Due to the enlargement of the cup, the mean gray-level of OD is found to be larger in a glaucoma eye as compared to the normal eye. In order to compute mean gray-level, primarily the input RGB image has been converted to gray scale image. The mean gray level of image is then computed.

Gray-level run length (f_{56} - f_{62})

The OD of the normal eye has more rough texture as compared to OD of Glaucoma eye. The gray level runs of uneven texture are found to be longer than smooth texture image. In the proposed method, seven features from the run length matrix have been computed. Among these seven features, first five features have been introduced by Galloway³⁴ and the last two features have been proposed by Chu et al.³⁵ The extracted seven features are as following.

$$\text{Short runs emphasis} = \frac{1}{n_r} \sum_{i=0}^{G-1} \sum_{j=1}^R \frac{P(i,j)}{j^2} \quad (9)$$

$$\text{Long runs emphasis} = \frac{1}{n_r} \sum_{i=0}^{G-1} \sum_{j=1}^R P(i,j) j^2 \quad (10)$$

$$\text{Gray-level nonuniformity} = \frac{1}{n_r} \sum_{i=0}^{G-1} \left(\sum_{j=1}^R P(i,j) \right)^2 \quad (11)$$

$$\text{Run length nonuniformity} = \frac{1}{n_r} \sum_{j=1}^R \left(\sum_{i=0}^{G-1} P(i,j) \right)^2 \quad (12)$$

$$\text{Run percentage} = \frac{n_r}{n_p} \quad (13)$$

$$\text{Low gray-level run emphasis} = \frac{1}{n_r} \sum_{i=1}^{G-1} \sum_{j=1}^R \frac{P(i,j)}{i^2} \quad (14)$$

$$\text{High gray-level run emphasis} = \frac{1}{n_r} \sum_{i=0}^{G-1} \sum_{j=1}^R P(i,j) \cdot i^2 \quad (15)$$

In the above equations, the n_r and n_p is the total number of runs and total number of pixels in the image respectively. Similarly, the $P(i, j)$ is the run length matrix.

Segmentation-based fractal texture analysis (f_{62} - f_{110})

The overall OD structure complexity of normal eye is found to be more complex than glaucomatous eye due to the presence of blood vessels. The segmentation based fractal texture analysis (SFTA) algorithm proposed in Reference 28 has been used to analyze the overall complexity of OD structure. The algorithm first splits input image into a set of binary images using multi-level Otsu algorithm. Next, the border of each binary image has been extracted. Finally, the fractal dimension of each border image, total area of each binary image and mean gray level of corresponding gray levels of each binary image has been computed. In the proposed system, SFTA algorithm is used to compute 48 dimensions.

Brightness (f_{111})

The glaucomatous OD is found to be brighter due to the presence of enlarged cup. The default weighted factors for RGB to YUV transformation have been used to compute brightness.³⁶

Color moments (f_{111} - f_{117})

Color moments extract global image color and intensity based information. It can analyze color similarities between the images.³⁷ In the proposed system, six color moments have been computed considering mean and SD of the red, green, and blue channels of the image.³⁸

Super-pixels (f_{118})

In the glaucoma eye, the numbers of super pixels are usually greater than normal eye due to the widening of the cup. In the proposed method, any pixel having a value greater than or equal to the threshold value $T = 0.9$ are termed as superpixels in red channel of input image. All survived pixels are counted after the thresholding.

3.1.3 | Comparisons with already deployed techniques

The proposed system has been tested on two publically available datasets and one local dataset. Table 2 gives summary of dataset used for testing.

The hybrid structural feature set (HSF) and hybrid textural feature set (HTF) system were evaluated on a

TABLE 2 Datasets used for testing

Dataset	Total images	Resolution	Healthy	Glaucoma
RIM-ONE	455	Different resolutions	200	255
High-resolution fundus (HRF)	30	3504 × 2336	15	50
Local database (LD-50)	50	415 × 452	25	25

TABLE 3 Comparison of results with already deployed techniques on local dataset with 50 images

	Cross-validation	Specificity (%)	Sensitivity (%)	Accuracy (%)
Automated detection of glaucoma using structural and nonstructural features. ⁴⁹	2 Folds			
CDR based detection		91	93	92
Feature based detection		91	86	90
Combined		88	100	92
Glaucoma detection using CDR and ISNT rule ⁵⁰	NA	85	73	82
Proposed system	2 Folds			
Proposed structural features module		92	96	94
Proposed textural features module		92	92	92
Combined results		89	100	94

TABLE 4 Comparison of results with already deployed techniques on HRF image dataset

	Data-set split ratio	Specificity (%)	Sensitivity (%)	Accuracy (%)
Glaucoma detection using hybrid feature set ⁵¹	60:40	86	93	90
Glaucoma diagnosis using retinal blood vessels ⁵²		93	80	86
Glaucoma detection by segmenting the super pixels from fundus color retinal images ⁵³		NA	NA	90
Proposed fundus module	60:40			
Proposed structural features module		93	73	83
Proposed textural features module		86	93	90
Combined results		86	100	94

TABLE 5 Comparison of Results with Already Deployed Techniques on RIM-ONE dataset

	Cross validation	Specificity (%)	Sensitivity (%)	Accuracy (%)
2D EWT and correntropy based glaucoma detection (455 images) ¹⁰	2 folds	84	76	80
Haralick features based glaucoma detection (321/455 images) ⁵⁶	2 folds	99	90	96
Polar map representation of the optic disc for glaucoma detection (100/455 images) ⁵⁷	NA	84	82	83
CDR based glaucoma detection (290/455 images) ¹²	NA	99	94	98
Proposed Findus module (455 images)	2 folds			
Proposed structural features module		79	71	76
Proposed textural features module		90	81	86
Combined results		80	91	85

publically available databases of 50 fundus images composed from MESSIDOR, HEI-MED, DRIVE, STARE, and DiaRetDB0. The comparison of the proposed fundus module with the algorithm in References 49 and 50 on a local dataset of 50 images is shown in Table 3. It can be clearly seen that proposed fundus module has outperformed both of the existing systems in term of accuracy and sensitivity.

Similarly, the performance of the proposed fundus module has been also compared with the techniques for glaucoma detection proposed by Akram et al,⁵¹ Manjesh and Nagachandr,⁵² and Dutta et al⁵³ on the publicly available HRF Image Database.⁵⁴ The proposed HTF module attained good results with 90% accuracy on the HRF image database. The combined result attained 94% accuracy and 100% sensitivity. The comparison of the proposed fundus module with the algorithm in References 51-53 has been shown in Table 4.

The performance results of proposed system on RIM-ONE database⁵⁵ are listed in Table 5. Only one of the existing method has used all 455 images of database for evaluation of system and delivered 80% accuracy. While the other three approaches have discarded noisy images and used selected images for testing of the methods. The proposed system has used all 455 images for the performance analysis and attained 85% accuracy, 80% specificity, and 91% sensitivity on RIM-ONE database.

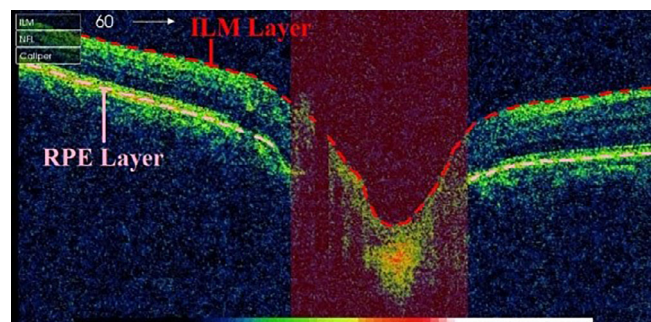


FIGURE 7 Retinal layers used for CDR computation [Color figure can be viewed at wileyonlinelibrary.com]

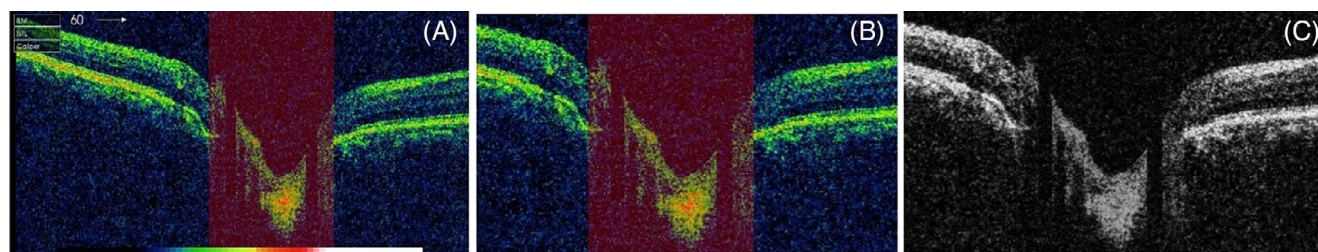


FIGURE 8 Green channel, (A) original image, (B) manually cropped and resized image (s-image), and (C) green channel (g-image) of s-image [Color figure can be viewed at wileyonlinelibrary.com]

3.2 | OCT module

In the OCT module, two most prominent retinal layers have been used for cup and disc extraction. The ILM layer has been used for cup edges detection. Whereas, the RPE layer has been used for the disc edges detection. Figure 7 highlights ILM and RPE layers on a SD-OCT image. The green channel of SD-OCT image has been used for both cup and disc extraction. The green channel gives a vivid view of retinal layers by eliminating the effects of red translucent ribbon generated during OCT scan.

Figure 8A shows the original image captured during OCT scan. The spatial resolution has been increased using bilinear interpolation before region of interest segmentation. During segmentation, the cup is always placed exactly in the center with equal retinal layers on both sides as shown in Figure 8B. The green channel of scaled s-image can be seen in Figure 8C.

3.2.1 | Thresholding strategy

In the proposed system, we have computed a set of threshold values based on input image gray level distribution by using multi-level Otsu algorithm.³⁹ The threshold value, which minimizes the input image intra-class variance has been selected. The Otsu algorithm is then recursively applied to each image region until the desired number of threshold n_t is obtained. In the proposed system, we have computed six-level threshold values by taking $n_t = 6$ and selected different levels to segment disc and cup from the input image.

3.2.2 | The ILM layer extraction

De-noising and gray scale conversion

The SD-OCT images encounter speckle noise due to the coherent property of image construction process. Wiener filter can reduce speckle noise using 2D adaptive noise removal filtration.⁴⁰ It filters an image based

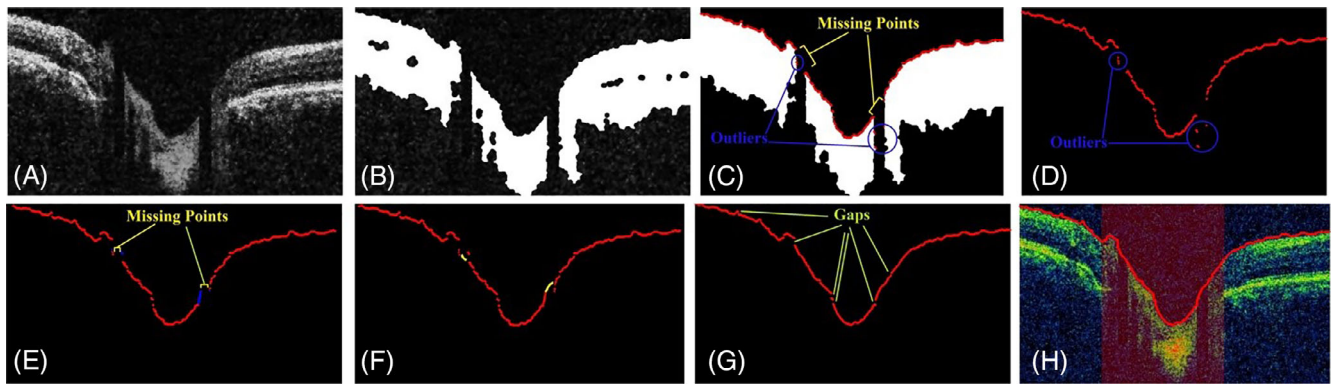


FIGURE 9 ILM layer extraction, (a) grayscale image (gs-image) of g-image after speckle noise removal, (b) boosted ILM layer after ILM layer filtration process, (c) binary image (ILM-b-image) after morphological operations, (d) extracted top surface of ILM layer (TS_{ILM}) plotted in red color on top of ILM layer, (e) missing points on TS_{ILM} shown in yellow color, (f) shows interpolated points in yellow color and highlights outliers and gaps on TS_{ILM} , (g) refine TS_{ILM} drawn on a blank binary image (TS-image) using proposed methods, (h) refine TS_{ILM} plotted on s-image [Color figure can be viewed at wileyonlinelibrary.com]

on neighborhood pixels by computing mean and SD.⁴¹ In ILM layer extraction, 15×15 neighborhood pixels have been used to filter g-image using wiener filter. The grayscale conversion has been done after the speckle noise removal. Figure 9A shows gs-image after the speckle noise elimination.

ILM-layer filtration

The accurate ILM layer extraction is essential for accurate cup diameter calculation. In ILM layer filtration process, a unique method has been introduced to enhance ILM layer without boosting the compressed speckle noise. Primarily, the gs-image (Figure 9A) is split into another gs-image image in the ILM layer filtration process, that is, gs-image1. The gs-image1 has been obtained by scaling the gs-image by a factor of 2. This operation will make lesser visible portion of ILM layer brighter and conspicuous as compared to the background. On the other hand, it will also boost compressed speckle noise once again. The contrast of the ILM-layer has been further adjusted so that 1% of the data is saturated at low and high intensities.³⁰ The enhanced ILM layer is then extracted using level-3 threshold value. The boosted speckle noise is then removed by removing any object smaller than P pixels. Where P is derived from average size of ILM layer in given dataset. Some disconnected smaller portions of ILM layer can also be removed during the removal of boosted speckle noise. This ILM layer loss has been handled by merging initial gs-image with gs-image1 (Figure 9B). The Figure 9B shows boosted ILM layer after ILM layer filtration process.

Image enhancement and thresholding

The contrast of merged image has been once again enhanced for better segmentation results. Next, morphological closing

using a disk-shaped structuring element with a radius of 10 has been applied to fuse together the extracted ILM layer portions from both grayscale images. Remaining gaps and holes have been further filled using morphological operation to get a fully merged ILM layer. After morphological operations, the level 4 threshold value has been used to segment ILM layer. Noise removal has been done by removing any object smaller than P pixels. Figure 9C shows ILM-b-image with extracted ILM layer after thresholding.

ILM layer top surface extraction and refinement

The TS_{ILM} has been computed by identifying row index of first ON pixel in each column of b-image.⁴² Figure 9D shows extracted TS_{ILM} plotted in red color on the top of ILM layer. In some cases, the extracted TS_{ILM} encounters some outliers representing extraction error. The presence of outlier can mislead the interpretation of gathered data.⁴³ The outliers are detected in the proposed system based on distance-based outlier detection method. After detection, the outlier has been reallocated to the best possible location considering the overall trend of the data. The Figure 10A-1 shows an exemplary TS_{ILM} with outliers highlighted in red squares. The proposed technique to remove outliers calculates mean value of left set and right set neighboring ON pixels and then averaging mean value of both side determine the new location for detected outlier. The neighboring pixels involved in reallocation of outlier can be visualized in Figure 10B-1. For simulation simplification, the minimum acceptable distance is set to 1 ($D = 1$). Any point having a distance greater than D will be considered as an outlier and will be reallocated to a new position by considering neighboring pixel. The colored square around four neighbors on the both sides of each outlier shows eight neighboring points involved in assigning a new position to outlier shown in the same

color. The dull black points show original position of the outliers. In Figure 9D, some outliers encircled in blue color can be observed. It can be clearly seen that the outliers portion do not conform to the overall trends of TS_{ILM} data. In the proposed system, the minimum acceptable distance is taken as 5 while 70 immediate left and right neighboring pixels are considered to allocate a new position to outlier point. In Figure 9E, it can be clearly seen that the proposed outlier removal method has effectively identified outliers and reallocate them considering the general fashion of TS_{ILM} .

The extracted TS_{ILM} is then analyzed for any missing point. A missing point is generated in case an ON pixel is absent in any column. These missing points are again predicted by averaging method using a set of left and right ON pixels. Figure 10A-2 shows an exemplary TS_{ILM} with some missing points. Figure 10B-2 simulates the proposed interpolation strategy. In Figure 10B-1, colored squares on both sides of a specific point indicates set of left and right neighboring points involved in the prediction of a missing point shown in the same color. The proposed interpolation method predicted the most appropriate points in the negative, positive and flat slope of the line. Figure 9E shows extracted TS_{ILM} with some missing portion of ILM layer. The interpolated missing points using proposed interpolation method can be visualized in Figure 9F shown in yellow color. After the outliers removal, it has been required to fill gaps on TS_{ILM} for accurate cup edges determination. A gap occurs due to the empty rows between two consecutive points on TS_{ILM} . Figure 10A-3 shows TS_{ILM} with no missing points and outliers; however, it has multiple gaps. The TS_{ILM} holds row position of each "ON" pixel, the difference between two consecutive points on TS_{ILM} can be used to check the number of blank rows between two points. If the difference is greater than 2, then

it indicates a gap and the absolute value of difference also specifies the number of missing rows in a gap. The Figure 10B-3 shows strategy adopted to fill the gaps. In the proposed method, initially the TS_{ILM} has been drawn on the blank binary image (TS-image) of the same size as b-image. Then the gaps are identified and filled using the proposed method. A fully solid TS_{ILM} drawn on TS-image can be seen in Figure 9G. The accuracy of extracted and refined TS_{ILM} can be visualized in Figure 9H.

3.2.3 | The RPE layer extraction

Contrast enhancement, de-noising, and thresholding

In the process of RPE layer extraction, the speckle noise has been again removed by means of wiener filter using 10×10 structuring element. Figure 11A shows noise free green channel. Similarly, Figure 11B shows grayscale image after contrast enhancement using histogram equalization and morphological operation. The morphological closing using a disk-shaped structuring element with a radius of 15 has been applied to merge together all small portions of ILM layer and RPE layer and also to isolate both layers. The level 5 threshold value has been used to segment the retinal layers. The binary image after thresholding and removing objects smaller than P pixels is shown in Figure 11C.

ILM layer extraction and removal

An innovative strategy has been adopted to remove ILM and connected layers on top of the RPE layer based on the already calculated TS_{ILM} . According to clinical investigations, the quantitative measure of thickness profile of six retinal layers on the top of RPE-layer in a normal healthy subject is $200 \mu m$ on SD-OCT images.⁴⁴ The standard

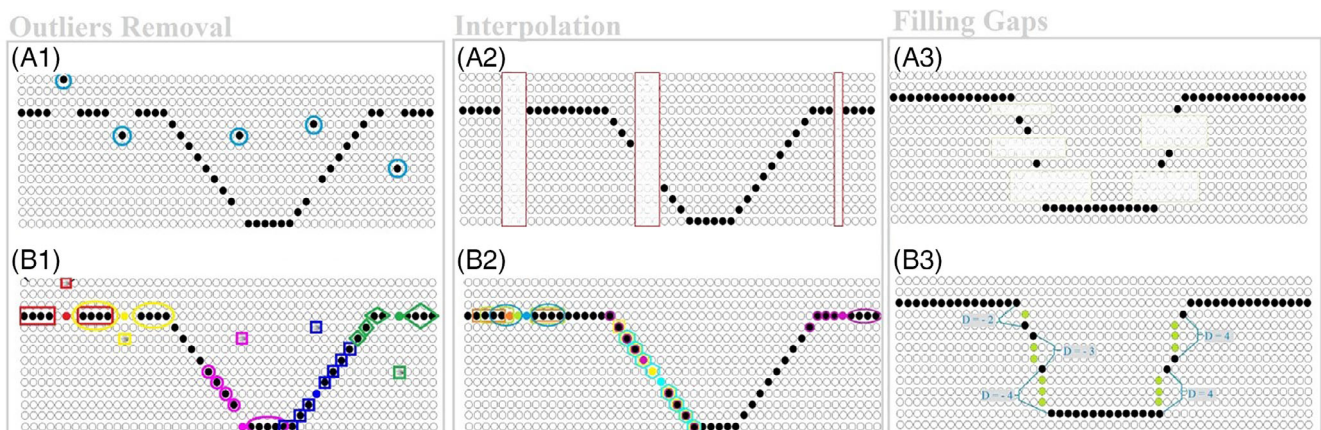


FIGURE 10 TS_{ILM} refinement method simulation, (a-1) TS_{ILM} with some outlier highlighted in blue squares, (b-1) the proposed strategy to remove outliers, (a-2) TS_{ILM} with highlighted missing points, (b-2) the proposed strategy to fill missing points (a-3) TS_{ILM} with highlighted gaps, (b-3) the proposed strategy to fill gaps [Color figure can be viewed at wileyonlinelibrary.com]

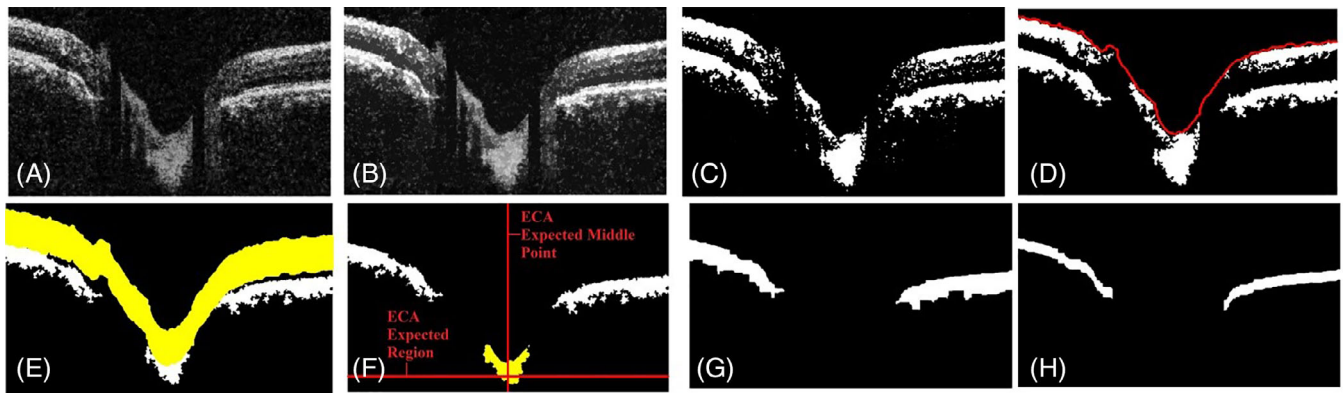


FIGURE 11 RPE layer extraction, (A) green channel after noise removal, (B) grayscale conversion and morphological operations, (C) binary image (RPE-b-image) after thresholding, (D) Already extracted TS_{ILM} plotted on RPE-b-image, (E) the yellow portion highlights retinal layers to be removed on top of RPE layer on the bases of already calculated TS_{ILM} , (F) the vertical red line shows the ECA expected middle point, the highlighted area beneath horizontal red line is ECA expected region, the yellow portion is the identified ECA to be remove, (G) RPE and brunch membrane (BM) layer after ECA removal (RPE-image), and (H) precise extraction of RPE layer (PRPE-image) [Color figure can be viewed at wileyonlinelibrary.com]

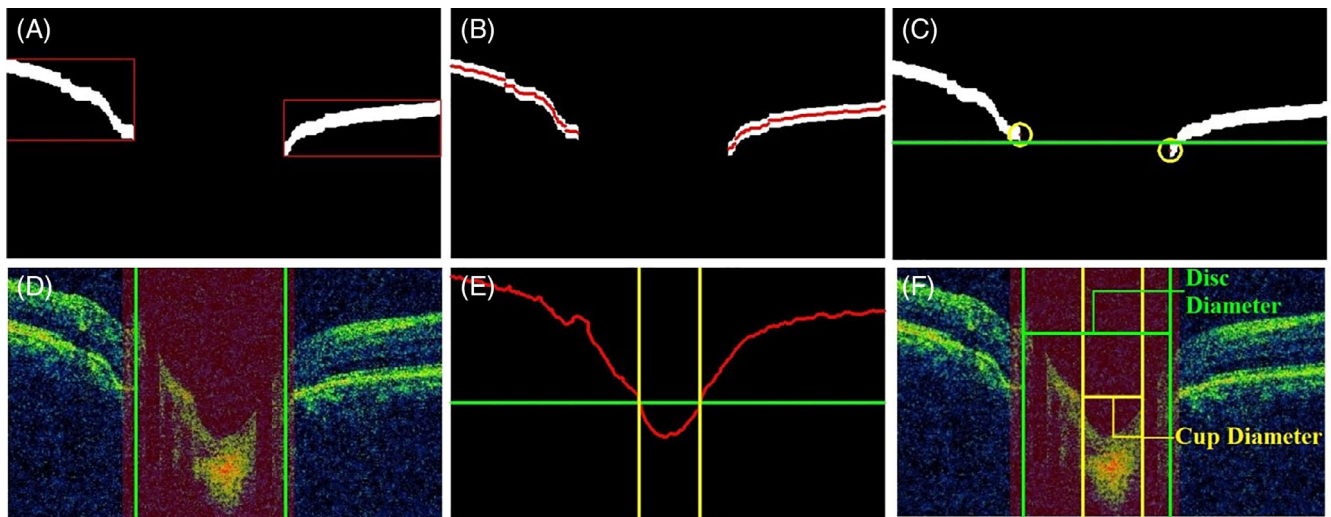


FIGURE 12 CDR calculation, (a) RPE layer region analysis, (c) the red line shows calculated middle points of RPE layer, (d) identified disc edges based on RPE layer end points, (e) identified cup edges based on mean value of RPE layer end points on TS-image, and (f) cup and disc diameter calculated using proposed method [Color figure can be viewed at wileyonlinelibrary.com]

SD-OCT image has an axial resolution of $3 \mu\text{m}/\text{pixel}$. Since we have scaled images and enlarged them three times than their original size; therefore, we have used ILM-layer thickness-value $T_{ILM} = 200$ pixels to remove maximum layers on the top of RPE-layer. In the proposed ILM removal method, a portion of ILM layer is computed first using TS_{ILM} calculated in CDC process and an empirically calculated ILM Layer thickness value T_{ILM} . Later, the calculated portion has been removed to uncover RPE layer by removing all layers on top of it. Figure 11D shows already calculated TS_{ILM} plotted in the red color on RPE-b-image. Similarly, ILM layer portion to be removed using TS_{ILM} and T_{ILM} is shown in yellow color in Figure 11E.

RPE layer end points calculation and CDR computation

The extended cup areas (ECA) are those parts of cup that persist during ILM layer elimination. It is important to remove ECA for accurate and precise RPE layer extraction. The ECA removal has two main phases. In the first phase, a seed point is identified by searching ECA at specified areas. If ECA exists then region-growing algorithm is used to find all connected components of identified seed point.⁴⁵ Finally, all the connected components associated with the seed point are removed. Figure 11F shows ECA expected regions. Any object at these locations is considered as ECA and removed. The yellow portion shows the detected ECA to be removed based on the

seed point. Figure 11G shows RPE and BM layer after ECA removal. After ECA removal, the precise RPE layer extraction has been done by calculating thickness profile (TP) of extracted RPE layer. In RPE layer TP calculation, the top surface of extracted RPE layer (TS_{RPE}) in RPE-image is calculated by identifying first “ON” pixel in each column.⁴¹ The TP is then calculated by counting the number of all connected “ON” pixels to TS_{RPE} in RPE-image. The number with maximum occurrence in TP is considered as thickness (TK) of RPE layer. The TK is further divided by 2 to remove the BM layer beneath the RPE layer. Finally, the RPE layer is drawn based on TS_{RPE} and TK on a blank binary image of same size as of RPE-b-image as shown in Figure 11H.

In few cases, more than two portions of RPE layer have been obtained due to some ECA portions that remain undetected in ECA removal process. The RPE layer region analysis has been done to check the number of portions of RPE layer. The “regionprops” method has

been used to count the numbers of extracted RPE layer portions. Figure 12A shows the identified portion of RPE layer in red boxes during region analysis process. In case, more than two portions of RPE layer are identified then each RPE layer portion is cropped automatically using “regionprops” method; and size of each portion is obtained by counting a number of “ON” pixels in each portion. Finally, the two portions with maximum size are selected as valid RPE layer portions discarding other smaller portions. After extracting left and right RPE layer portions, the end points of both portions are computed. The process of RPE layer end point determination involves calculation of bottom surface of RPE layer (BS_{RPE}) by identified first ON pixel from bottom. Furthermore, the TK value is further divided by two to get the middle of RPE layer. BS_{RPE} is then drawn exactly in the center of RPE layer using TK as shown in Figure 12B. Once the middle of the RPE layer is identified then the BS_{RPE} is searched for first “OFF” pixel from both sides. Figure 12C shows the first “OFF”

TABLE 6 Classification results of different modules of H-CAD system [Color Table can be viewed at wileyonlinelibrary.com]

R	GT	HSF	HTF	OM	H-CAD	L	GT	HSF	HTF	OM	H-CAD
R1	N	G	N	N	N	L1	N	N	N	N	N
R2	G	G	N	G	G	L2	G	G	N	G	G
R3	G	G	G	G	G	L3	N	N	G	N	N
R4	N	N	N	N	N	L4	N	G	N	N	N
R5	N	G	N	G	G	L5	N	N	N	N	N
R6	G	N	G	G	G	L6	G	G	G	G	G
R7	G	G	G	N	S	L7	N	N	N	N	N
R8	G	N	N	G	S	L8	G	N	G	G	G
R9	N	G	N	N	N	L9	N	G	N	N	N
R10	G	N	G	G	G	L10	G	G	G	G	G
R11	G	G	G	G	G	L11	N	N	N	N	N
R12	G	G	G	G	G	L12	G	G	G	G	G
R13	N	N	N	N	N	L13	G	G	G	G	G
R14	G	G	G	G	G	L14	G	G	G	G	G
R15	N	G	G	N	S	L15	N	N	N	N	N
R16	N	N	N	N	N	L16	N	G	N	N	N
R17	G	G	G	G	G	L17	G	G	G	G	G
R18	G	G	G	G	G	L18	G	G	G	G	G
R19	G	G	N	G	G	L19	G	G	G	G	G
R20	G	G	G	N	S	L20	G	G	G	G	G
R21	G	G	G	G	G	L21	G	G	G	G	G
R22	N	N	N	N	N	L22	N	N	G	N	N
R23	N	G	N	N	N	L23	N	N	N	N	N
R24	G	G	G	G	G	L24	G	G	G	G	G
R25	N	N	G	N	N	L25	N	N	N	N	N

TABLE 7 HSF module performance

	Healthy	Glaucoma
Healthy	14	8
Glaucoma	4	24

TABLE 8 HTF module performance

	Healthy	Glaucoma
Healthy	18	4
Glaucoma	4	24

pixels encircled in yellow color. These points are considered as RPE layer end points and cup and disc edges are finally determined on the basis of these termination points of RPE layer.

According to studies, the termination points of the RPE-layer are not likely to change substantially with glaucoma progression, and can serve as a stable reference plane.^{46,47} The cup at a level of the RPE-layer in the SD-OCT images correlates well with clinical reference standards.⁴⁸ During our analysis, the mean value of the RPE-layer end points as a reference plane provided better correlation with clinical results to detect the cup edges. The RPE layer end points define the disc edges and the distance between both end points is taken as disc diameter. Figure 12D shows identified disc margins using RPE layer end points. Similarly, the cup edges are also detected based on RPE layer end points. In cup edges determination, the mean value of RPE layer end points is computed first. In Figure 12C, the green line represents the mean value of the RPE layer end points. Next, cup edges are located on TS-image based on the mean value as shown in Figure 12E. The distance between cup edges is taken as cup diameter. Figure 12F shows calculated disc and cup diameters. In case, no cup edges are detected in between RPE layer end points then the cup is considered as flat with an absolute CDR = 0.2. The CDR value is then computed by taking percentage of cup diameter to the disc diameter.

4 | RESULTS

4.1 | Data set description

The system has been evaluated on a custom dataset acquired from Armed-Forces-Institute-of-Ophthalmology (AFIO). The dataset consists of 50 fundus and 50 SD-OCT images comprising left and right eye images of 25 patients with 951×456 and 2032×1934 resolutions.

TABLE 9 OCT module performance

	Healthy	Glaucoma
Healthy	21	1
Glaucoma	2	26

TABLE 10 H-CAD system performance

	Healthy	Glaucoma	Suspected
Healthy	20	1	1
Glaucoma	0	25	3

Twenty-two images in each set have been of healthy eye, whereas 28 images have been of glaucoma eyes. Each patient data has been blindly tested by four different ophthalmologists with varying experience. The ophthalmologist 1 is the junior most having least experience whereas; the ophthalmologist 4 is the senior most with maximum experience.

4.2 | Results

Table 6 shows classification results of HSF, HTF, OCT module (OM), and also fused classification result of H-CAD system. The ground truth (GT) represents final result of AFIO dataset. The proposed fundus module has shown significantly improved results as compared to the other automated systems evaluated on some other publicly available dataset (Tables 3–5). In case of the classification of the results using custom dataset acquired from AFIO, a reduction in overall accuracy of fundus module can be seen. The reduced accuracy, especially in case of the HTF module is due to the extremely poor visualization of the fundus images. It was almost impossible to locate the cup and disc on these low quality images. However, the OCT images of the same patients have been able to generate accurate results after the application of the proposed layer enhancement methods.

The HSF module detected 38 out of 50 fundus images accurately with 76% accuracy (Inaccurate results are highlighted with green color in Table 6). The performance of the HSF module can further be observed in Table 7. It can be seen that the HSF module has been able to detect 14 out of 22 normal cases with 64% specificity and 24 out of 28 glaucoma cases have been identified correctly with 86% sensitivity.

Whereas the HTF module achieved relatively better performance with 42% accurately classified images out of 50 fundus images with 84% accuracy (Inaccurate results are highlighted with yellow color in Table 6). The performance

TABLE 11 The final result of H-CAD system

	Accuracy (%)	Specificity (%)	Sensitivity (%)
HSF module	76	64	86
HTF module	84	82	86
OCT module	94	95	93
H-CAD system	96	91	100

TABLE 12 Dataset summary for clinical evaluation

Images	Resolution	H	G	Image type
66	951 × 456	32	34	SD-OCT
66	2032 × 1934	32	34	Fundus

of the HTF module can also be analyzed in Table 8. It has achieved 82% specificity with 18 correctly classified images out of 22 normal cases. It has also attained 86% sensitivity by correctly classifying 24 out of the 28 glaucoma cases.

Excellent results have been obtained in the case of classification using the OCT images. The OCT module has been able to accurately detect 47 out of 50 images with 94% accuracy (Inaccurate result is highlighted with blue color in Table 6). Then, 93% sensitivity has been achieved by correctly classifying 26 out of 28 glaucoma cases. Only one normal image has been classified as glaucoma attaining 95% specificity as shown in Table 9.

The result after fusing results of HSF, HTF and OCT modalities can be analyzed by observing the H-CAD results in Table 6. An obvious improvement in the overall result can be seen with 100% sensitivity and 96% accuracy. Table 10 shows the performance of H-CAD system in detail. The basic motivation behind developing the proposed H-CAD system was to involve capabilities of both imaging modalities for the accurate and reliable diagnosis of glaucoma.

The final result determination of the H-CAD system is based on three completely different modules. These modules are capable of analyzing OD in entirely different ways to ensure more reliable glaucoma diagnosis than its contemporary automated models. Table 11 shows an overview of the performance of the H-CAD system.

4.3 | Comparison of proposed H-CAD system with clinical results

The proposed H-CAD system is under clinical trials from last 6 months. Table 12 gives summary of dataset used for clinical evaluation. The results of the proposed H-CAD system are quite satisfactory. The results have shown

TABLE 13 Comparison of proposed H-CAD system results with clinical results

	Healthy	Glaucoma
(a): OPT1 result		
Healthy	23	8
Glaucoma	7	28
(b): OPT2 result		
Healthy	26	3
Glaucoma	5	32
(c): OPT3 result		
Healthy	32	4
Glaucoma	2	28
(d): OPT4 result		
Healthy	29	3
Glaucoma	2	32

TABLE 14 Performance of proposed H-CAD system with clinical results

	OPT1 (%)	OPT2 (%)	OPT3 (%)	OPT4 (%)
Sensitivity	80	86	93	94
Specificity	74	90	88	91
Accuracy	77	88	91	92

more correlation with senior ophthalmologist's result. The result analysis of four ophthalmologists can be seen in Table 13. The H-CAD system achieved 77% accuracy with OPT1, 88% with OPT2, 91% with OPT3, and 92% accuracy with OPT4 as shown in Table 14. These results demonstrate that the proposed system can be used as a decision support system to assist junior doctors.

5 | DISCUSSION

Medical images can play a significant part to improve public health for all age groups. It can produce images of internal structures of the body without surgical interventions. Glaucoma is an asymptomatic disease at the initial and treatable stage. It shows the symptoms in the last stages after irreversible vision loss and ultimately results in permanent blindness. The regular screening of patient can detect this chronic disease at an early and curable stage which may avoid the permanent loss of sight.

The proposed H-CAD system can harness this silent thief of sight by considering a number of glaucoma indicated by scanning the outer and internal layers of the retina using fundus and OCT imaging technology. The fundus and OCT image analysis is equally important for

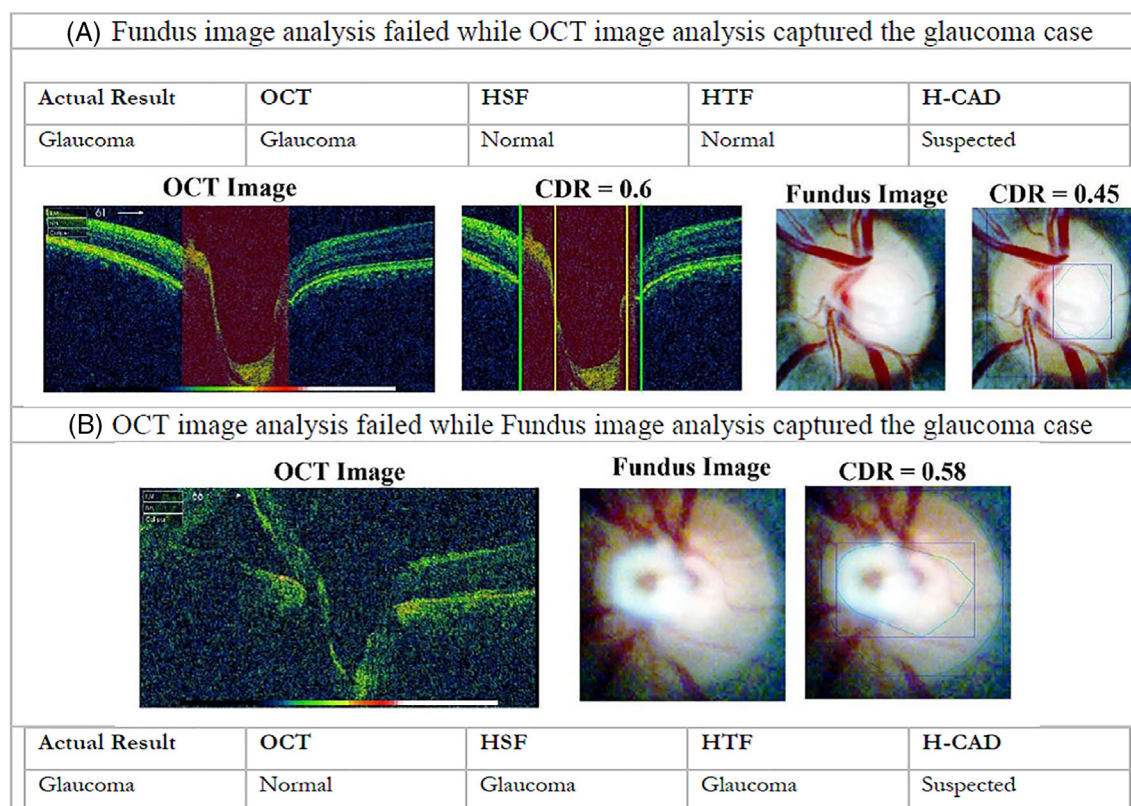


FIGURE 13 Importance of fusion of fundus and OCT image analysis [Color figure can be viewed at wileyonlinelibrary.com]

complete and through analysis of glaucoma. During evaluation it was observed that in cases where fundus image analysis failed to sense glaucoma the OCT image analysis was able to detect changes in internal layers of retina. However, on the contrary such case was also encountered where the OCT image analysis failed due to poor visualization of retinal layers and intense speckle noise, but the fundus image analysis played a vital role and detected the glaucoma based on overall appearance and structural changes occurred due to glaucoma. Figure 13 illustrates such cases with results.

Generally, the automated glaucoma diagnosis has been accomplished by using CDR in the fundus imaging modality and few have used features along with classification. The SD-OCT is an emerging optical technology in medical imaging. It is an under researched area and relatively minimum research work has been done for automated diagnosis of glaucoma in this field. The great potential of this technology motivated us to vigorously research the utility of OCT in the detection of glaucoma. In the proposed system, we have integrated the powers of both imaging modalities for more accurate, reliable and early diagnosis of glaucoma. In fundus module, three structural features that is, RDR, CDR, and cup shape have been extracted. These three structural features enable the system to analyze OD structure in several ways. Similarly, 115 textural features have been

extracted by assessing the overall appearance and texture of OD. The global texture and intensity based information of the complete OD portion has been computed. Moreover, the texture information of subparts of OD portion including rim vessels, frequency details in multiple orientations and analysis of cup using proposed super-pixel calculation method has been extracted. In this way, the proposed fundus module has been able to generate improved classification results as compared to the existing CAD system. None of the existing systems has ever considered such a large and versatile combination of different structural and textural features for glaucoma diagnosis. The classification has been performed on the HTF and HSF feature separately and the support vector machine (SVM) has been used as the classifier. The SVM is famous for best handling of multidimensional data, binary classification and robustness added with the help of different kernels.⁵⁸ In our case, we have used radial basis function (RBF) as kernel along with SVM. The RBF kernel is capable of yielding higher accuracy as compared to linear and third degree polynomial kernel function on multidimensional data.⁵⁹ The involvement of two separate classifiers for the categorization and inclusion of suspected class has played an important part in the overall enhanced result of fundus module.

In the OCT module, the CDR is computed considering internal layers of the retina. The cup contour has

extracted from ILM-layer. The unique strategy to boost the ILM-layer has given accurate results even with extreme low visualization of ILM-layers. Likewise, the novel methods to refine ILM-layer contour has predicted the most appropriate values for missing points in the case of low-quality images. Similarly, for accurate RPE-layer extraction, a unique strategy to remove all upper layers on the top of RPE-layer has given excellent results for RPE layer extraction in all cases. Furthermore, the technique to compute RPE-layer thickness and precisely extraction of RPE-layer based on thickness value has given most appropriate points to compute cup diameter. The new criterion to locate cup edges has given remarkably improved and accurate results in the case of CDR calculation for both the normal and glaucomatous eyes.

6 | CONTRIBUTIONS

The contributions of the study are summarized as follows:

1. Gathering and annotation of a database containing the fundus and OCT images for each patient. There is no online available dataset which provides information for combined analysis of the OCT and fundus images. The developed dataset has been made available on request on our research group.*
2. A glaucoma detection system has been proposed by combining and correlating the analysis of the fundus and OCT images.
3. A unique set of structural and textural features has been proposed from the ONH part of the fundus image considering a variety of glaucoma conditions.
4. A new method has been introduced to improve the CDR calculation from the fundus images considering two different color channels.
5. A new technique has been proposed for reliable extraction of the ILM-layer in the low quality SD-OCT images.
6. A new strategy has been presented to remove all upper layers on top of the RPE-layer for accurate extraction and thickness calculation of the RPE-layer in the SD-OCT images.

7 | CONCLUSION

The fusion of classification results of both modules has given improved results with 96% accuracy and 100% accuracy for glaucoma referral. The improved result of H-CAD system demonstrates the effectiveness and power of integration of both modalities for the accurate and reliable diagnosis. This rigorous analysis of internal and outer retinal layer ensures that no glaucoma affected case remains

undiagnosed. Hence, in rural areas where there is a scarcity of ophthalmologist, the proposed system can be used for glaucoma diagnosis and only genuine cases can be forwarded to the specialists for further examinations.

In future, this research can be extended for inclusion of the clinical indicators such as intra ocular pressure (IOP) and visual field (VF) measurements along with the image analysis for more clinical correlations and improved results. The functional analysis of the eye on the above mentioned indicators will further add diversity in the analysis process. The early diagnosis and prediction of disease can be achieved in future by analysis of macula and ganglion cells along with the ONH region analysis. One of the early indicators of glaucoma is the degeneration of ganglion cells. The total macula thickness loss can be used as an indicator of ganglion cells. The decay of ganglion cell can also be analyzed by measuring thickness of the Retinal Nerve Fiber (RNFL) layer. The 3D generation of the RNF-layer for thickness estimation can be an excellent indicator to diagnose glaucoma at an early and treatable stage. Another way to effectively diagnose this disease could be by correlating both eyes of patient to analyze symmetrical patterns as the ISNT measures of both eyes are not symmetrical in the glaucoma.

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CONFLICT OF INTEREST

All the authors of this study declare that there is no conflict of interests regarding the publication of this manuscript.

ETHICAL APPROVAL

This article does not contain any studies with human participants or animals performed by any of the authors.

ORCID

Tehmina Shehryar  <https://orcid.org/0000-0002-7674-1468>

ENDNOTE

*website www.biomisa.org

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